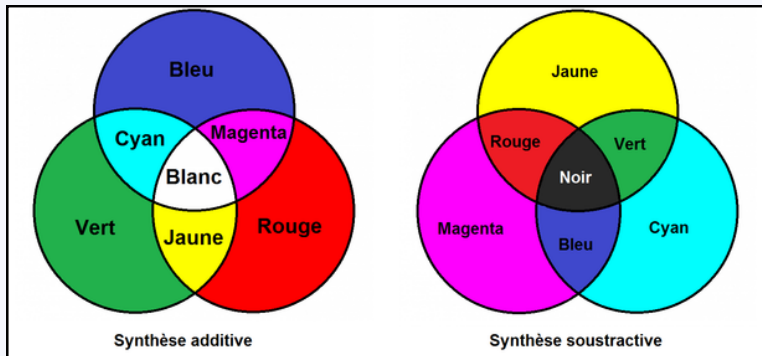
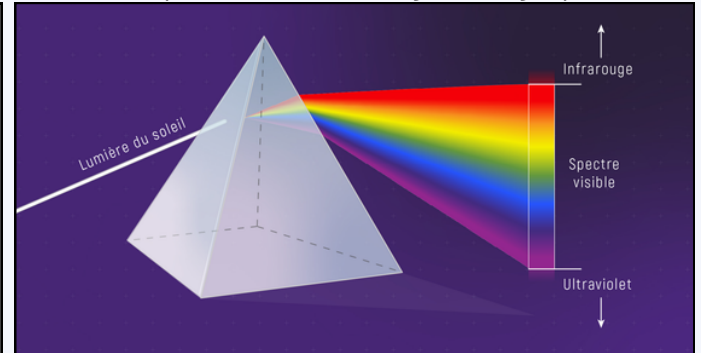


TECHNICAL SHEET (FOR EXPERTS / TEACHERS) – WORKSHOP 2 : GEOMETRIC & CHROMATIC OPTICS

Color Circles



Decomposition of white light using a prism



Background : This educational workshop focuses on the spectral composition of white light and the mechanisms of human color perception.

Objectives of the experiment : To decompose polychromatic white light through a dispersive medium (prism), and to recreate white light, as well as shadows of secondary colors, through additive synthesis.

I. Physical Phenomena in Focus

- Light dispersion (differential refraction) : When polychromatic white light passes through a K9 crystal glass prism, each wavelength (λ) is deflected by a different amount. This phenomenon is governed by **Snell-Descartes' law** : $n_1 \cdot \sin(\theta_1) = n_2(\lambda) \cdot \sin(\theta_2)$
- The refractive index of the glass, $n_2(\lambda)$, depends on the wavelength (Cauchy's law). Blue ($\lambda \simeq 450$ nm) is deflected more than red ($\lambda \simeq 650$ nm), which separates the visible spectrum.
- Additive synthesis (RGB model) : superposition of light fluxes from three primary wavelengths (Red, Green, Blue). The simultaneous stimulation of the three types of cones in the human retina is interpreted by the visual cortex as white light.
- Induced polarization and colored shadows : when an opaque obstacle blocks one of the beams (e.g., red), the geometric shadow cast by that source receives only the residual light fluxes (green + blue), creating a shadow of the complementary secondary color (cyan).

II. Experimental Protocol and Technical Alignment

1. Light beam generation : Adjust the zoom of the white LED lamp to obtain a collimated beam (as straight as possible).
2. Dispersion : Position the K9 prism at the angle of minimum deviation to maximize the spatial separation of the rainbow on the screen.
3. RGB superposition : Position the three colored lamps so that their radiation patterns coincide at a single focal point on the wall, in order to achieve a perfect chromatic white.

DID YOU KNOW THAT ?

Did you know that ? The magic of screens and colorful shadows ! White light isn't a color in itself, but the combination of all the colors in the visible spectrum ! That's the principle of **additive color mixing (RGB)**. That's exactly how your smartphone screen works : it uses millions of microscopic red, green, and blue pixels to recreate all images. In our workshop, when your hand blocks the red light, the wall continues to receive green and blue light. That's why your shadow instantly turns **cyan** ! This concept of physical interaction and human perception is crucial in engineering for designing display and augmented reality technologies (Academic sources: IEEE Journal of Display Technology / IEEE Photonics Journal).